

CBSE Class 12
Computer Science
Previous Year Question Paper 2017

Series: GBM

Set- 4

Code no. 91

- Please check that this question paper contains **20** printed pages.
- Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **7** questions.
- Please write down the Serial Number of the question before attempting it.
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

COMPUTER SCIENCE

Time Allowed: 3 hours

Maximum Marks: 70

General Instructions:

- SECTION A** refers to the programming language C++.
- SECTION B** refers to the programming language Python.
- SECTION C** is compulsory for all.
- Answer either **SECTION A** or **SECTION B**.
- It is compulsory to mention on page 1 in the answer book whether you are attempting **SECTION A** or **SECTION B**.
- All questions are compulsory within each section.

SECTION - A

Only for candidates, who opted for C++

1. (a) Write the type of C++ tokens (keywords and user defined identifiers) from the following: 2 Marks

- (i) New
- (ii) While
- (iii) Case
- (iv) Num_2

Ans: (i) New - Keyword

(ii) While - User defined Identifier

(iii) Case - Keyword

(iv) Num_2- User defined Identifier

(b) Anil typed the following C++ code and during compilation he found three errors as follows: 1 Mark

(i) Function strlen should have prototype

(ii) Undefined symbol cout

(iii) Undefined symbol endl

On asking, his teacher told him to include necessary header files in the code. Write the names of the header files, which Anil needs to include, for successful compilation and execution of the following code:

```
void main()
{
    char Txt[] = "Welcome";
    for (int C= 0; C<strlen (Txt); C++)
        Txt [c] = Txt[c] + 1;
    Cout<<Txt<<end;
}
```

Ans: string.h

iostream.h OR fstream.h OR iomanip.h

(c) Rewrite the following C++ code after removing any/all syntactical errors with each correction underlined. **2 Marks**

Note: Assume all required header files are already being included in the program.

```
Void main ()
{
cout<<"Enter an Alphabet:"
cin>>CH;
switch (CH)
    case 'A' cout<<"Ant"; Break;
    case 'B'cout<<"Bear"; Break;
}
```

Ans: void main ()

```
{
cout<<"Enter an Alphabet:";
charCH; // Error 1
cin>>CH;
switch(CH)
{ //Error 2 (i)
case 'A': // Error 3 (i)
    cout<<"Ant"; break; // Error 4 (i)
case 'B': // Error 3 (i)
    cout<<"Bear"; break; // Error 4(ii)
} // Error 2(ii)
}
```

(d) Find and write the output of the following C++ program code: **2 Marks**

Note: Assume all required header files are already included in the program.

```
#define Diff (N1, N2) ((N1>N2)? N1-N2:N2-N1)
```

```
void main ()
```

```
{
    int A,B,NUM[] = {10, 23, 14, 54, 32};
    for (int CNT =4; CNT>0; CNT--)
    {
        A=NUM [CNT];
        B=NUM [CNT-1];
        Cout<<Diff (A, B) <<'#';
    }
}
```

Ans: 22#40#9#13#

(e) Find and write the output of the following C++ program code: 3 Marks

Note: Assume all required header files are already being included in the program.

```
void main ()
{
    int *Point, Score[]={100,95,150,75,65,120};
    Point = Score;
    for (int L = 0; L<6; L++)
    {
        If ((*Point) % 10==0)
            *Point /= 2;
        else
            *Point -= 2;
        If ((*Point) %5==0)
            *Point /= 5;
        Point++;
    }
    for (int L = 5; L>=0; L--)
        cout<<Score[L]<<"*";
```

}

Ans: 12*63*73*15*93*10*

(f) Look at the following C++ code and find the possible output(s) from the options (i) to (iv) following it. Also, write the maximum values that can be assigned to each of the variables N and M. **2 Marks**

Note:

- Assume all the required header files are already being included in the code.
- The function random(n) generates an integer between 0 and n – 1

```
void main()
{
    Randomize ();
    int N=random(3),M=random(4);
    int DOCK[3][3] = {{1,2,3},{2,3,4},{3,4,5}};
    for (int R=0; R<N; R++)
    {
        for (int C=0; C<M; C++)
            cout<<DOCK[R][C]<<" ";
        cout<<endl;
    }
}
```

(i)	(ii)
1 2 3 2 3 4 3 4 5	1 2 3 2 3 4
(iii)	(iv)
1 2 2 3	1 2 2 3 3 4

Ans: Correct Options: (ii) and (iii)

Maximum value of N = 2

Maximum value $M = 3$

2. (a) Differentiate between protected and private members of a class in context of Object Oriented Programming. Also give a suitable example illustrating accessibility/non-accessibility of each using a class and an object in C++.
2 Marks

Ans:

Private	Protected
Implicit Visibility Mode. Not accessible to member functions of derived class.	Explicit Visibility Mode. Accessible to member functions of derived class.

Example:

Class A

{

 Int x;

Protected:

 Int y;

Public:

 Void z ();

};

(b). Observe the following C++ code and answer the questions (i) and (ii).

Note: Assume all necessary files are included.

Class TEST

{

long TCode;

char TTitle[20];

float Score;

public:

TEST ()

{

```

Tcode=100;
Strcpy (TTitle,"FIRST Test");
Score=0;
}
TEST (TEST &T);
{
TCode=E.TCode+1;
Strcpy (TTitle,T.TTitle);
Score=T.Score;
}
void main()
{
    _____ //Statement 1
    _____ //Statement 2
}

```

(i) Which Object Oriented Programming feature is illustrated by the Member Function 1 and the Member Function 2 together in the class TEST?
1 Mark

Ans: Polymorphism OR Constructor overloading OR Function Overloading

(ii) Write Statement 1 and Statement 2 to execute Member Function 1 and Member Function 2 respectively.
1 Mark

Ans: TEST T2 (T1); **//Statement 1**
TEST T1 (T1); **//Statement2**

(c) Write the definition of a class BOX in C++ with the following description:
4 Marks

Private Members

- **BoxNumber** **// data member of integer type**
- **Side** **// data member of float type**
- **Area** **// data member of float type**
- **ExecArea ()** **// Member function to calculate and assign**

// Area as Side * Side

Public Member

- **GetBox ()** // A function to allow user to enter values of
 // BoxNumber and Side. Also, this
 // function should call ExecArea() to calculate
 // Area
- **ShowBox ()** // A function to display BoxNumber, Side
 // and Area

Ans: Class box

```
{
    Int BoxNumber;
    float side;
    float Area;
    void ExecArea() {
        Area=side*side;
    }
    public:
    void Getbox();
    void showbox;
};
void Box: :GetBox()
{
    cin>>BoxNumber>>Side
    ExecArea ();
}
Void Box : : ShowBox()
{
    cout<<BoxNumber<<" "<<Side<<" "<<Area<<endl;
```



```
}
```

(d) Answer the questions (i) to (iv) based on the following:

4 Marks

Class First

```
{
```

```
    int X1;
```

```
protected:
```

```
    float X2;
```

```
public:
```

```
    First();
```

```
    void Enter1(); void Display1();
```

```
};
```

Class Second : private First

```
{
```

```
    int Y1;
```

```
protected:
```

```
    float Y2;
```

```
public:
```

```
    Second();
```

```
    void Enter2();
```

```
    void Display();
```

```
};
```

class Third : public Second

```
{
```

```
    int Z1;
```

```
public:
```

```
    Third();
```

```
    void Enter3();
```

```
    void Display();
```

```
};  
void main()  
{  
    Third T;           //Statement 1  
    _____;       //Statement 2  
}
```

(i) Which type of Inheritance out of the following is illustrated in the above example? Single Level Inheritance, Multilevel Inheritance, Multiple Inheritance

Ans: Multilevel Inheritance

(ii) Write the names of all the member functions, which are directly accessible by the object T of class Third as declared in main () function.

Ans: Enter2 (), Display () of class Second
Enter3 (), Display () of class Third

(iii) Write Statement 2 to call function Display () of class Second from the object T of class Third.

Ans: T.Second : : Display ();

(iv) What will be the order of execution of the constructors, when the object T of class Third is declared inside main ()?

Ans: First, Second, Third

3. (a) Write the definition of a function AddUp (int Arr[], int N) in C++, in which all even positions (i.e., 0,2,4,...) of the array should be added with the content of the element in the next position and odd positions (i.e., 1,3,5,...) elements should be incremented by 10. 3 Marks

Example: if the array Arr contains

23	30	45	10	15	25
----	----	----	----	----	----

Then the array should become

53	40	55	20	40	25
----	----	----	----	----	----

Note:

- The function should only alter the content in the same array.

- The function should not copy the altered content in another array.
- The function should not display the altered content of the array.
- Assuming, the Number of elements in the array are Even

Ans: void AddUp (int Arr[], int N)

```
{
    for (int i=0; i<N;i++)
    {
        if (i%2==0)
            Arr[i] = Arr[i] + Arr[i+1];
        Else
            Arr[i] = Arr[i] + 10;
    }
}
```

(b) Write a definition for a function SUMMIDCOL (int MATRIX [][][10], int N,int M) in C++, which finds the sum of the middle column's elements of the MATRIX (Assuming N represents number of rows and M represents number of columns, which is an odd integer). Example: If the content of array MATRIX having N as 5 and M as 3 is as follows: 2 Marks

1	2	1
2	1	4
3	4	5
4	5	3
5	3	2

The function should calculate the sum and display the following:

Sum of Middle Column: 15

Ans: void SUMMIDCOL (int MATRIX [][] [10],int N,int M)

```
{
    int mid=M/2;
    int sum=0;
    for (int i=0; i<N; i++)
```

```
{
    sum = sum+MATRIX[i][mid];
}
cout<<"sum of middle column"<<sum;
}
```

(c) ARR [15] [20] is a two-dimensional array, which is stored in the memory along the row with each of its elements occupying 4 bytes. Find the address of the element ARR [5] [15], if the element ARR [10] [5] is stored at the memory location 35000. 3 Marks

Ans: ROW MAJOR

$\text{Loc (ARR [I] [J])} = \text{BaseAddress} + W [(I - \text{LBR}) * C + (J - \text{LBC})]$

(where W=size of each element = 4 bytes, R=Number of Rows=15, C=Number of Columns=20)

Assuming LBR = LBC = 0

LOC (ARR [10] [5])

$35000 = \text{BaseAddress} + W (I * C + J)$

$35000 = \text{BaseAddress} + 4 (10 * 20 + 5)$

$35000 = \text{BaseAddress} + 4 (205)$

$35000 = \text{BaseAddress} + 820$

$\text{BaseAddress} = 35000 - 820 = 34180$

$\text{LOC (ARR [5] [15])} = \text{BaseAddress} + W (I * C + J)$

$= 34180 + 4(5 * 20 + 15)$

$= 34180 + 4(100 + 15)$

$= 34180 + 4 \times 115$

$= 34180 + 460$

$= 34640$

(d) Write the definition of a member function PUSHGIFT() for a class STACK in C++, to add a GIFT in a dynamically allocated stack of GIFTS considering the following code is already written as a part of the program:

4 Marks

struct GIFT

```
{
int GCODE;                //Gift Code
char GDESC [20];          //Gift Description
GIFT *Link;
};
Class STACK
{
    Gift *TOP;
public:
    STACK () {
TOP= NULL;
}
    void PUSHGIFT ();
    void POPGIFT();
    STACK ();
};
```

Ans: void STACK::PUSHGIFT ()

```
{
GIFT *T = new GIFT;
cin>>T->GCODE;
gets (T->GDESC);
T->Link = TOP;
TOP = T;
}
```

(e) Convert the following Infix expression to its equivalent Postfix expression, showing the stack contents for each step of conversion: 2 Marks

$X - (Y + Z) / U * V$

Ans:

Element	Stack	Postfix
x		X
	-	X
(	-(	X
Y	-(	XY
+	-(+	XY
Z	-(+	XYZ
)	-	XYZ+
/	-/	XYZ+
U	-/	XYZ+U
*	-*	XYZ+U/
v	-*	XYZ+U/V
		XYZ+U/V*

4. (a) Polina Raj has used a text editing software to type some text in an article. After saving the article as MYNOTES.TXT, she realised that she has wrongly typed alphabet K in place of alphabet C everywhere in the article.

3 Marks

Write a function definition for PURETEXT () in C++ that would display the corrected version of the entire article of the file MYNOTES.TXT with all the alphabets "K" to be displayed as an alphabet "C" on screen.

Note: Assuming that MYNOTES.TXT does not contain any C alphabet otherwise.

Example:

If Polina has stored the following content in the file MYNOTES. TXT:

I OWN A KUTE LITTLE KAR.
I KARE FOR IT AS MY KHILD.

The function PURETEXT () should display the following content:

I OWN A CUTE LITTLE CAR.
I CARE FOR IT AS MY CHILD.

Ans: void PURETEXT ()

```
{
char ch;
```

```
ifstream F("MYNOTES.TXT" );
while (F.get(ch))
{
    if (ch=='K')
    ch='C';
    cout<<ch;
}
F.close ();      //IGNORE
}
```

(b) Write a definition for function COUNTPICS() in C++ to read each object of a binary file PHOTOS.DAT, find and display the total number of PHOTOS of type PORTRAIT. Assume that the file PHOTOS.DAT is created with the help of objects of class PHOTOS, which is defined below:

2 Marks

```
class PHOTOS
{
    Int PCODE;
    Char PTYPE [20];      //Photo Type as "PORTRAIT", "NATURE"
public:
    void ENTER ()
    {
        cin>>PCODE;gets (PTYPE);
    }
    void SHOWCASE ()
    {
        cout<<PCODE<<":" <<PTYPE<<endl;
    }
    char *GETPTYPE(){
return PTYPE;
}
```

```
};
```

Ans: void COUNTPICS ()

```
{
ifstream F;
F.open ("PHOTOS.Dios : : binary);
int count=0;
PHOTOS obj;
while (F.read ((char*)&obj,sizeof (obj)))
{
if (strcmp(obj.GETPTYPE(),"PORTRAIT")==0)
Count ++;
}
cout<<"Number of PORPRAIT photos : "<<count;
F.close ();      //IGNORE
}
```

(c) Find the output of the following C++ code considering that the binary file CLIENTS.DAT exists on the hard disk with a data of 200 clients: 1 Mark

```
class CLIENTS
{
    int CCode; char CName[20];
public:
    void REGISTER(); void DISPLAY();
};

void main()
{
    fstream File;
    File.open("CLIENTS.DAT",ios::binary|ios::in);
    CLIENTS C;
    File.seekg(6*sizeof(C));
```



```
File.read((char*)&C, sizeof(C));
cout<<"Client Number:"<<.tellg () /sizeof (c) +1;
File.seekg(0,ios: :end);
cout<<" of "<<File.tellg () /sizeof(c)<<endl;
File.close();
}
```

Ans: Client Number 8 of 200

SECTION - B

Only for candidates, who opted for Python

1. (a) Which of the following can be used as valid variable identifier(s) in Python? 2 Marks

- (i) 4thSum
- (ii) Total
- (iii) Number#
- (iv) _Data

Ans: (ii) Total (iv) _Data

(b) Name the Python Library modules which need to be imported to invoke the following functions: 1 Mark

- (i) floor ()
- (ii) randint ()

Ans: Math

Random

(c) Rewrite the following code in Python after removing all syntax error(s). Underline each correction done in the code. 2 Marks

STRING=""WELCOME

NOTE""

for S in range[0,8]:

print STRING(S)

print S+STRING

Ans: STRING="WELCOME"

NOTE=" "

for S in range(0,8): print STRING[S] print S,STRING Also range(0,8) will give a runtime error as the index is out of range. It should be range (0, 7)

(d) Find and write the output of the following Python code: 2 Marks

TXT = ["20","50","30","40"]

CNT = 3

TOTAL = 0

for C in [7,5,4,6]:

T = TXT [CNT]

TOTAL = float (T) + C

print TOTAL

CNT-=1

Ans: 47.0

35.0

54.0

26.0

(e) Find and write the output of the following Python code: 3 Marks

class INVENTORY:

def __init__(self,C=101,N="Pad",Q=100): #constructor

self.Code=C

self.IName=N

self.Qty=int(Q);

def Procure(self,Q):

self.Qty = self.Qty +

def Issue(self,Q):

self.Qty -= Q

def Status(self):

```
print self.Code,":",self.IName,"#",self.Qty
```

```
I1=INVENTORY ()
```

```
I2=INVENTORY (105,"Thumb Pin", 50)
```

```
I3=INVENTORY (102,"U Clip")
```

```
I1. Procure (25)
```

```
I2. Issue (15)
```

```
I3. Procure (50)
```

```
I1. Status ()
```

```
I3. Status ()
```

```
I2. Status ()
```

Ans: Output

101: Pad # 125

102: U Clip # 150

105: Thumb Pin # 35

(f) What are the possible outcome(s) executed from the following code?

2 Marks

```
for C in range (NUM, 1, -1):
```

Also specify the maximum and minimum values that can be assigned to the variable N

```
import random
```

```
NAV = ["LEFT","FRONT","RIGHT","BACK"];
```

```
NUM = random.randint (1, 3)
```

```
NAVG = ""
```

```
NAVG = NAVG+ NAV [I]
```

```
print NAVG
```

(iii) BACK	(iv) LEFTFRONTRIGHT
(i) BACKRIGHT	(ii) BACKRIGHTFRONT

Ans. (i) BACKRIGHT

Max value 3 and minimum value 1 for variable NUM

2. (a) List four characteristics of Object Oriented Programming. 2 Marks

Ans. Four characteristics of Object Oriented Programming are:

- Encapsulation
- Data Hiding
- Inheritance
- Polymorphism

(b) class Exam:

Regno=1

Marks=75

def __init__(self,r,m): #function 1

self.Regno= r

self.Marks= m

def Assign(self,r,m): #function 2

Regno = r

Marks = m

def Check (self): #function 3

print self.Regno, self.Marks

print Regno, Marks

(i) In the above class definition, both the functions — function 1 as well as function 2 have similar definition. How are they different in execution?

1 Mark

Ans: Function 1 is the function Object () {[native code]}, which is called automatically when the class object is created. Function 2 is a member function that must be invoked in order to assign values to Regno and Marks.

(ii) Write statements to execute function 1 and function 2.

1 Mark

Ans: Function 1 E1= Exam (1, 95) #Any values in the parameter

Function 2 E1.Assign (1, 95) #Any values in the parameter

(c) Define a class BOX in Python with the following specifications: 4 Marks

Instance Attributes

- BoxID

Numeric value with a default value 101

- ## Methods:

Dynamic binding: It allows you to link a function while it's running. That is, until the function that is to be connected with the function call is executed, the code of the function that is to be linked with the function call is unknown. Programs are more versatile when functions are dynamically bound.

(e) Write two methods in Python using the concept of Function Overloading (Polymorphism) to perform the following operations: 2 Marks

(i) A function having one argument as Radius, to calculate Area of Circle as $3.14 \times \text{Radius} \times \text{Radius}$.

(ii) A function having two arguments as Base and Height, to calculate Area of right-angled triangle as $0.5 \times \text{Base} \times \text{Height}$.

Ans: def Area (R):

print $3.14 \times R \times R$

def Area (B,H):

print $0.5 \times B \times H$

3. (a) What will be the status of the following list after the First, Second and Third pass of the bubble sort method used for arranging the following elements in ascending order? 3 Marks

Note: Show the status of all the elements after each pass very clearly underlining the changes.

52, 42, -10, 60, 90, 20

Ans: I Pass

52	42	-10	60	90	20
42	52	-10	60	90	20
42	-10	52	60	90	20
42	-10	52	60	90	20
42	-10	52	60	90	20
42	-10	52	60	90	90

II pass

42	-10	52	60	20	90
-10	42	52	60	20	90
-10	42	52	60	20	90
-10	42	52	60	20	90
-10	42	52	20	60	90

III Pass

-10	42	52	20	60	90
-10	42	52	20	60	90
-10	42	52	20	60	90
-10	42	52	52	60	90

(b) Write definition of a method EvenSum(NUMBERS) to add those values in the list of NUMBERS, which are odd. 3 Marks

Ans: def EvenSum (NUMBERS):

n=len (NUMBERS)

s=0

for i in range(n)

: if (i%2!=0):

s= s+NUMBERS[i]

print (s)

(c) Write Addnew (Member) and Remove (Member) methods in Python to Add a new Member and Remove a Member from a list of Members, considering them to act as INSERT and DELETE operations of the data structure Queue. 4 Marks

Ans: class queue:

Member= []

def Addnew(self):

a=input ("enter member name: ")

queue.Member.append (a)

def Remove (self):

if (queue.Member==[]):

print "Queue empty"

else:

print "deleted element is: ",queue.Member[0]

del queue.Member[0] # queue.Member.delete()

(d) Write definition of a method MSEARCH (STATES) to display all the state names from a list of STATES, which are starting with alphabet M.

2 Marks

For example:

If the list STATES contains

["MP","UP","WB","TN","MH","MZ","DL","BH","RJ","HR"]

The following should get displayed:

MP

MH

MZ

Ans: def MSEARCH (STATES):

for i in STATES:

if i [0]=='M':

print i

(e) Evaluate the following Postfix notation of expression:

2 Marks

4, 2, *, 22, 5, 6, +, /, -

Ans:

ELEMENT	Stack Contents
4	4
2	4 ,2
*	8
22	8 , 22
5	8, 22, 5
6	8 , 22, 5, 6
+	8 , 22 ,11
/	8 , 2
-	6

Answer: 6

4. (a) Differentiate between file modes r+ and rb+ with respect to Python.

1 Mark

Ans. Opens a file for both reading and writing with the r+ command. The file pointer is positioned at the start of the file.

rb+ opens a file in binary format for both reading and writing. The file pointer is positioned at the start of the file.

(b) Write a method in Python to read lines from a text file MYNOTES.TXT, and display those lines, which are starting with the alphabet —"K". 2 Marks

Ans: def display ():

```
file=open('MYNOTES.TXT','r')
line= file.readline()
while line:
    if line[0]=='K' :
        print line
    line= file.readline()
file.close ()           #IGNORE
```

(c) Considering the following definition of class FACTORY, write a method in Python to search and display the content in a pickled file FACTORY.DAT, where FCTID is matching with the value "105". 3 Marks

Class Factory:

```
def __init__(self,FID,FNAM):
    self.FCTID = FID           # FCTID Factory ID
    self.FCTNM = FNAM         # FCTNM Factory Name
    self.PROD = 1000          # PROD Production

def Display (self):
    print self.FCTID,":",self.FCTNM,":", self.PROD
```

Ans: import pickle

```
def ques4c ( ):
    f=Factory ( )
    file =open('FACTORY.DAT','rb')
    try:
        while True:
```

```
f=pickle.load (file)
if f.FCTID==105:
    f.Display ()
except EOF Error:
    pass
file.close()          #IGNORE
```

SECTION - C

For all the candidates

5. (a) Observe the following table **MEMBER** carefully and write the name of the RDBMS operation out of (i) **SELECTION** (ii) **PROJECTION** (iii) **UNION** (iv) **CARTESIAN PRODUCT**, which has been used to produce the output as shown in **RESULT**. Also, find the Degree and Cardinality of the **RESULT**:
2 Marks

MEMBER

NO	MNAME	STREAM
M001	JAYA	SCIENCE
M002	ADITYA	HUMANITIES
M003	HANSRAJ	SCIENCE
M004	SHIVAK	COMMERCE

RESULT

NO	MNAME	STREAM
M002	ADITYA	HUMANITIES

Ans: SELECTION

Degree= 3

Cardinality=1

(b) Write SQL queries for (i) to (iv) and find outputs for SQL queries (v) to (viii), which are based on the tables. 6 Marks

DVD

DCODE	DTITLE	DTYPE
F101	Henry	FOLK
C102	Dhrupad	Classical
C101	The planets	Classical
F102	Universal Soldier	Folk
R102	A day in life	Rock

MEMBER

MID	NAME	DCODE	ISSUE DATE
101	AGAM SINGH	R102	2017-11-30
103	ARTH JOSEPH	F102	2017-11-30
102	NISHA HANS	C101	2017-07-24

(i) To display all details from the table MEMBER in descending order of ISSUEDATE.

Ans: SELECT * FROM MEMBER ORDER BY ISSUEDATE DESC;

(ii) To display the DCODE and DTITLE of all Folk Type DVDs from the table DVD.

Ans: SELECT DCODE, DTITLE FROM DVD WHERE DTYPE='Folk';

(iii) To display the DTYPE and number of DVDs in each DTYPE from the table DVD.

Ans: SELECT COUNT (*), DTYPE FROM DVD GROUP BY DTYPE;

(iv) To display all NAME and ISSUEDATE of those members from the table MEMBER who have DVDs issued (i.e., ISSUEDATE) in the year 2017.

Ans: SELECT NAME, ISSUEDATE FROM MEMBER WHERE
ISSUEDATE>='2017-01-01' AND ISSUEDATE<='2017-12-31';

(v) SELECT MIN (ISSUEDATE) FROM MEMBER;

Ans: MIN (ISSUEDATE)

2016-12-13

(vi) SELECT DISTINCT DTYPE FROM DVD;

Ans: DISTINCT DTYPE

Folk

Classical

Rock

(vii) SELECT D.DCODE, NAME, DTITLE FROM DVD D, MEMBER M WHERE D.DCODE=M.DCODE;

Ans:

DCODE	NAME	DTITLE
R102	AGAM SINGH	A day in life
F102	ARTH JOSEPH	Universal Soldier
C101	NISHA HANS	The Planets

(viii) SELECT DTITLE FROM DVD WHERE DTYPE NOT IN ("Folk", "Classical");

Ans: DTITLE

6. (a) State DeMorgan's Laws of Boolean Algebra and verify them using truth table. 2 Marks

Ans: (i) $(A.B)' = A' + B'$

(ii) $(A+B)' = A'.B'$

Truth Table Verification:

(i)

A	B	A.B	$(A.B)'$	A'	B'	$A' + B'$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

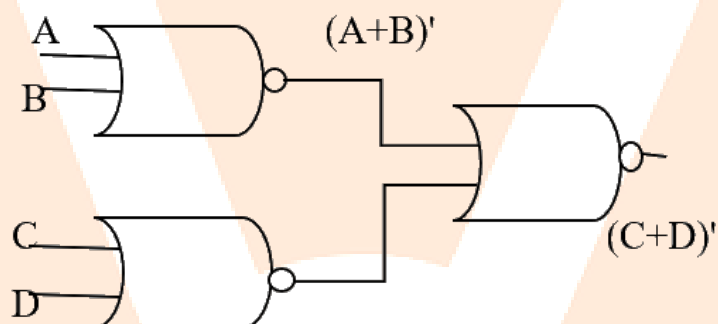
(ii)

A	B	A+B	(A+B)'	A'	B'	A'.B'
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

(b) Draw the Logic Circuit of the following Boolean Expression using only NOR Gates: 2 Marks

$$(A+B). (C+D)$$

Ans:



(c) Derive a Canonical POS expression for a Boolean function G, represented by the following truth table: 1 Mark

X	Y	Z	G(X,Y,Z)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Ans: $G(X, Y, Z) = (X+Y+Z). (X+Y+Z'). (X+Y'+Z'). (X'+Y'+Z)$

(d) Draw the Logic Circuit of the following Boolean Expression using only NOR Gates: **3 Marks**

E (U , V , Z , W) = $\Sigma (2 , 3 , 6 , 8 , 9 , 10 , 11 , 12 , 13)$

Ans: UV' U'V UV UV'

Z'W'			1	1
Z'W			1	1
ZW	1			1
ZW'	1	1		1

7. (a) Differentiate between communication using Optical Fiber and Ethernet Cable in context of wired medium of communication technologies. **2 Marks**

Ans: Optical Fibre

- Very Fast
- Expensive
- Immune to electromagnetic interference

Ethernet cable

- Slower as compared to Optical Fiber
- Less Expensive as compared to Optical Fiber
- prone to electromagnetic interference

(b) Janish Khanna used a pen drive to copy files from his friend's laptop to his office computer. Soon his computer started abnormal functioning. Sometimes it would restart by itself and sometimes it would stop different applications running on it. Which of the following options out of (i) to (iv), would have caused the malfunctioning of the computer? Justify the reason for your chosen option: **2 Marks**

(i) Computer Virus

(ii) Spam Mail

(iii) Computer Bacteria

(iv) Trojan Horse

Ans: Computer Virus

Justification:

- Pen drive containing Computer Virus was used before the abnormal functioning started, which might have corrupted the system files.
- Computer Virus effects the system files and start abnormal functioning in the computer

(c) Ms. Raveena Sen is an IT expert and a freelancer. She recently used her skills to access the Admin password for the network server of Super Dooper Technology Ltd. and provided confidential data of the organization to its CEO, informing him about the vulnerability of their network security. Out of the following options (i) to (iv), which one most appropriately defines Ms. Sen?

2 Marks

Justify the reason for your chosen option:

(i) Hacker

(ii) Cracker

(iii) Operator

(iv) Network admin

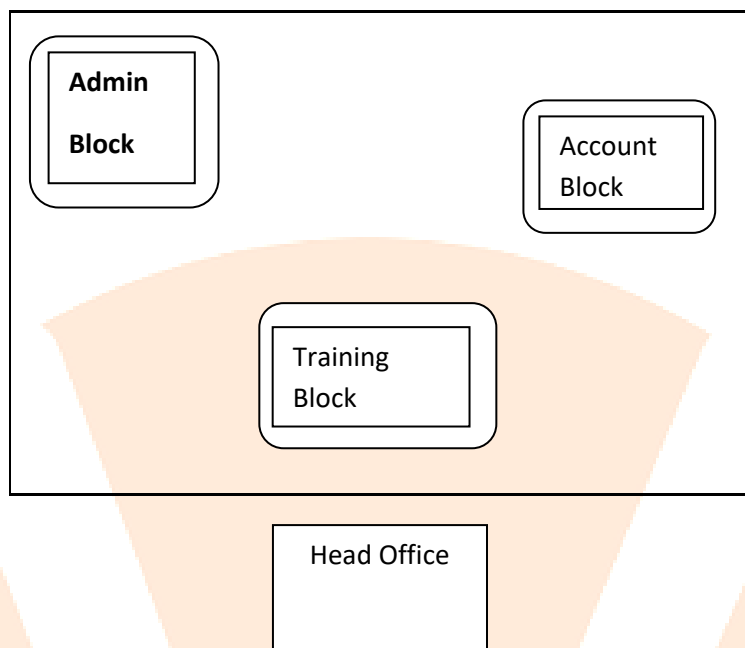
Ans: Hacker

A Hacker is a person who breaks into the network of an organization without any malicious intent.

(d) Hi Standard Tech Training Ltd. is a Mumbai based organization which is expanding its office set-up to Chennai. At Chennai office compound, they are planning to have 3 different blocks for Admin, Training and Accounts related activities. Each block has a number of computers, which are required to be connected in a network for communication, data and resource sharing.

As a network consultant, you have to suggest the best network related solutions for them for issues/problems raised by them in (i) to (iv), as per the distances between various blocks/locations and other given parameters.

CHENNAI Office



Shortest distances between various blocks/locations:

Admin Block to Accounts Block	300 Metres
Accounts Block to Training Block	150 Metres
Admin Block to Training Block	200 Metres
MUMBAI Head Office to CHENNAI Office	1300 Km

Number of computers installed at various blocks are as follows:

Accounts Block	150
Accounts Block	30
Admin Block	40

(i) Suggest the most appropriate block/location to house the SERVER in the CHENNAI office (out of the 3 blocks) to get the best and effective connectivity. Justify your answer. 1 Mark

Ans: Training Block - Because it has maximum number of computers.

(ii) Suggest the best wired medium and draw the cable layout (Block to Block) to efficiently connect various blocks within the CHENNAI office compound. 1 Mark

Ans: Best wired medium: Optical Fibre OR CAT5 OR CAT6 OR CAT7 OR CAT8 OR Ethernet Cable

(iii) Suggest the best wired medium and draw the cable layout (Block to Block) to efficiently connect various blocks within the CHENNAI office compound.

1 Mark

Ans: Firewall - Placed with the server at the Training Block

(iv) Suggest a device and the protocol that shall be needed to provide wireless Internet access to all smartphone/laptop users in the CHENNAI office.

1 Mark

Ans: Device Name: WiFi Router OR WiMax OR RF Router OR Wireless Modem OR RF Transmitter

Protocol: WAP OR 802.16 OR TCP/IP OR VOIP OR MACP OR 802.11